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Alert Prioritisation in Security Operations Centres: A Systematic Survey on Criteria and Methods

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Security Operations Centres (SOCs) are specialised facilities where security analysts leverage advanced technologies to monitor, detect and respond to cyber incidents. However, the increasing volume of security incidents has overwhelmed security analysts, leading to alert fatigue. Effective alert prioritisation (AP) becomes crucial to address this problem through the utilisation of proper criteria and methods. Human-AI teaming (HAT) has the potential to significantly enhance AP by combining the complementary strengths of humans and AI. AI excels in processing large volumes of alert data, identifying anomalies, uncovering hidden patterns, and prioritising alerts at scale, all at machine speed. Human analysts can leverage their expertise to investigate prioritised alerts, re-prioritise them based on additional context and provide valuable feedback to the AI system, reducing false positives and ensuring critical alerts are prioritised. This work provides a comprehensive review of the criteria and methods for AP in SOC. We analyse the advantages and disadvantages of the different categories of AP criteria and methods based on HAT, specifically considering automation, augmentation and collaboration. We also identify several areas for future research. We anticipate that our findings will contribute to the advancement of AP techniques, fostering more effective security incident response in SOCs.

CCS Concepts: • **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability;

Additional Key Words and Phrases: Security operations centre, alert prioritisation, criteria, method, human-AI teaming, automation, augmentation, collaboration

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1 Introduction

Cyber threats have surged in recent years, with global cybersecurity spending expected to rise from \$6 trillion to \$30 trillion by 2030 [124]. **Security Operations Centres (SOCs)** are becoming

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crucial for organizations to detect, investigate and respond to cyber threats effectively. SOC's can be viewed as a *human-machine team*, where a variety of AI-driven tools detect potential security incidents. Human analysts then investigate these incidents, assess their severity and take appropriate action to mitigate the threat [72]. However, the high volume of alerts overwhelms analysts, leading to fatigue, burnout and delayed responses to critical threats. Recent reports indicate that most organisations receive over 10,000 alerts daily, with more than 50% being false positives [32, 48]. In such cases, critical alerts may be missed among numerous less important or false-positive alerts, potentially causing significant harm to the organisation. Additionally, analysts feel overwhelmed and fatigued. They may also perceive their primary duty as reducing alert volume rather than investigating and addressing threats [90]. A 2021 study by Trend Micro [95] found that overwhelmed SOC analysts resort to turning off alerts, ignoring them or offloading them to colleagues.

Effective **alert prioritisation (AP)** is crucial for SOC's to effectively identify critical threats among a high volume of alerts, respond to them quickly and reduce analyst fatigue [90, 128]. Effective AP requires the use of proper criteria and methods. Different criteria can lead to varied identification and prioritization of critical alerts, significantly affecting SOC's' ability to respond promptly. For example, prioritising incidents using only their *frequency* (an alert-specific criterion) differs from using *asset criticality* or *asset replaceability* (asset-related criteria) as well. Additionally, the choice of AP method can greatly affect prioritisation accuracy and the SOC's' overall incident response efficiency. For instance, **machine learning (ML)** techniques facilitate processing of large data volumes for alert classification and prioritisation, while **multi-criteria decision-making (MCDM)** methods allow the inclusion of a wide range of criteria, with different weights (importance).

Human-AI teaming (HAT) harnesses the complementary strengths of both human and machine intelligence, allowing them to work together to achieve superior results that neither could achieve alone [29, 107]. The rapid advancements in AI have led to the incorporation of AI-powered technologies within SOC's, including **Security Information and Event Management (SIEM)** systems and **Extended Detection and Response (XDR)** platforms. SIEMs collect, correlate and analyse security event data from various sources, while XDRs combine multiple security controls and use advanced analytics to detect threats [19]. Despite the growing use of AI tools, studies like the SANS Institute's SOC Survey [33] and TrendMicro's Global Study [95] reveal that SOC's still face challenges affecting their efficiency in handling cyber threats. For example, the overwhelming volume of unprioritised alerts, the existence of many unintegrated tools and the lack of context for novel incidents hinder effective decision-making and response. Effective HAT can utilise the unique strengths of both human analysts and AI systems within SOC's while addressing the respective limitations of each. HAT has the potential to tackle complex and dynamic security situations in SOC's by combining human analysts' intuition, domain knowledge and expertise with AI's computational abilities and data processing power [129]. Accordingly, AP, crucial for determining alert criticality within the SOC, can gain significant benefits from HAT. AI can process large volumes of alert data, identify patterns and detect critical alerts that human analysts might overlook. Human analysts can then use their expertise to do in-depth analysis on AI-prioritised alerts, discover novel security incidents, re-prioritise based on additional context and offer valuable feedback to the AI.

HAT is typically realised in three ways: *automation*, *augmentation* and *collaboration* [19, 43, 129]. *Automation* uses technology to perform tasks that were previously done by humans. In this mode, AI independently manages decision-making without any human intervention. For example, by replacing human effort with machines for routine and well-known activities, automation enhances productivity and reduces costs by freeing up SOC analysts to focus on tasks demanding domain expertise [129]. *Augmentation* employs technology to improve or complement human decision-making, aiming to assist and optimise human performance without requiring direct collaboration.

Its primary objective is to augment the effectiveness of humans in their respective roles [40, 129]. It is a one-way relationship where the human remains the primary decision-maker benefiting from insights, explanations and recommendations provided by AI. For example, an ML technique can detect patterns in security alerts and provide explanations to analysts about the causes of security incidents, aiding them in preventing future incidents. *Collaboration* refers to dynamic, real-time and two-way interactions between humans and AI working to solve complex problems, make decisions or complete tasks that neither could achieve alone [19, 129]. In this mode, human experts tackle complex and uncertain issues by leveraging AI systems for collaborative exploration and investigation [19]. Collaboration allows humans and AI to enhance each other's capabilities. For example, in the dynamic and uncertain cybersecurity environment, analysts contribute their domain knowledge and expertise, while AI brings speed, memory capacity and computational capabilities in joint exploration and investigation of unknowns [19, 43, 129].

The three dimensions of HAT—*automation*, *augmentation* and *collaboration*—are expected to have distinct implications for AP criteria and methods. Automated AP methods can properly prioritise well-known and frequently occurring alerts using specific AP criteria. Augmented AP methods prioritise alerts and offer insights and explanations to analysts, enabling them to use a broader set of criteria when making final decisions on alert priorities. Collaborative AP methods are particularly effective in prioritising new, unknown and complex alerts by dynamically leveraging analysts' domain knowledge and AI capabilities in joint prioritisation.

This study reviews AP criteria and methods in SOC through the lens of HAT, focusing on automation, augmentation and collaboration. We use the method proposed by Wolfswinkel et al. [136] to systematically review the literature and answer the following three research questions:

- (i) *What are the main criteria used for AP in SOC?*
- (ii) *What are the main methods used for AP in SOC?*
- (iii) *How can HAT improve AP in SOC?*

To answer these questions, we review 89 papers on AP in SOC that were identified based on the review method described in Section 3. In doing so, we make the following main contributions:

- (i) The first systematic review of knowledge on AP within the SOC domain
- (ii) A taxonomy of AP criteria and methods in SOC
- (iii) Critical evaluation of the strengths and limitations of each category of criteria and method from the perspective of HAT
- (iv) Recommendations and future research directions to enhance AP in SOC by leveraging HAT

These contributions serve to advance the understanding and practice of AP in SOC, paving the way for improved alert management and cybersecurity operations through the integration of HAT.

Multiple papers review topics relevant to AP in SOC [2, 13, 63]. Hubballi and Suryanarayanan [63] reviewed false alert minimisation techniques in signature-based intrusion detection systems. Agalit et al. [2] reviewed techniques used for handling alerts generated by intrusion detection systems. Apruzzese et al. [13] reviewed the role of ML methods in cybersecurity tasks including alert management and risk assessment. To our knowledge, this is the first study to systematically categorise AP criteria and methods using the HAT lens. It provides a comprehensive understanding of the potential advantages of using AI for AP in SOC, including automation, augmentation and collaboration.

The article is organised as follows. Section 2 gives background on key concepts. Section 3 provides the review method. Sections 4 and 5 present the taxonomy of AP criteria and methods, respectively. Sections 6 and 7 discuss the strengths and limitations and future research directions of AP criteria and methods, respectively. Section 8 proposes additional future research directions

for AP in SOC. Section 9 outlines challenges for researchers and developers. Section 10 concludes the article.

2 Background

This section offers a succinct introduction to SOC and related concepts, providing essential background knowledge to understand the problem of AP in SOC.

2.1 Security Operations Centre

SOC is a specialised facility where a team of security analysts leverages a wide range of technologies, including AI-enabled systems, in conjunction with well-established procedures. The primary objective of SOC is to manage cybersecurity incidents and protect an organisation's IT assets from cyber attacks [72]. The SOC's core responsibilities include (1) prevention of security incidents, (2) thorough real-time alert detection and prioritisation, (3) thorough investigation of potential incidents, (4) prompt response to verified incidents, and (5) regularly reporting security status and incident outcomes to stakeholders [72]. By fulfilling these tasks, SOCs maintain the security posture of organisations and ensure the resilience of their IT infrastructure.

2.2 Alert Prioritisation

A typical SOC collects, analyses and stores between millions and billions of security events daily. A *security event* refers to “any observable occurrence in the system and/or network” [72]. To facilitate this process, SOC teams employ advanced tools like SIEM to generate security alerts. A *security alert* serves as “a technical notification that a particular event, or series of events, has occurred” [72].

AP in SOC involves sorting and organising alerts generated by diverse security tools, such as **Intrusion Detection Systems (IDSs)** and SIEM. The primary objective of AP is to assess the relative importance of alerts and to present the most critical ones to analysts for validation and investigation [72]. Alerts can sometimes be *false positives*, mistakenly indicating a threat where none exists. A cyber threat is any incident resulting in “an actual or potentially adverse effect” on the system [72]. Following AP, analysts validate alert priorities and investigate critical alerts. Figure 1 shows the process of AP in SOC. The use of advanced technologies and tools enables a relatively small group of analysts to effectively handle alerts derived from millions of security events.

Both humans and AI play key roles in AP [24, 72]. For example, the SIEM generates, enriches and prioritises alerts before presenting them to security analysts. Analysts then can validate alerts in the system's order or use their discretion to set the order. Once the analysts decide the priority order, they apply their tacit knowledge, experience or any additional contextual knowledge to decide whether to further investigate the alert, in which case it gets escalated, or take no further action if it is deemed a false positive or a benign alert [4].

The interplay between human analysts and AI in AP warrants HAT, which offers the potential for significantly improved outcomes. To achieve this, HAT can leverage three modes of decision-making: automation, augmentation and collaboration, as depicted in Figure 1.

- *Automation*: Automation can efficiently handle a large volume of alerts, reducing the initial burden on human analysts. In automated AP, the AI performs the task without analyst involvement, applying specific criteria and methods for AP. Following AP, the steps of alert validation are also fully automated, with the analysts not undertaking any investigation or validation tasks on alerts. The main drawback of this approach is that the machine can miss or misinterpret unknown critical alerts.

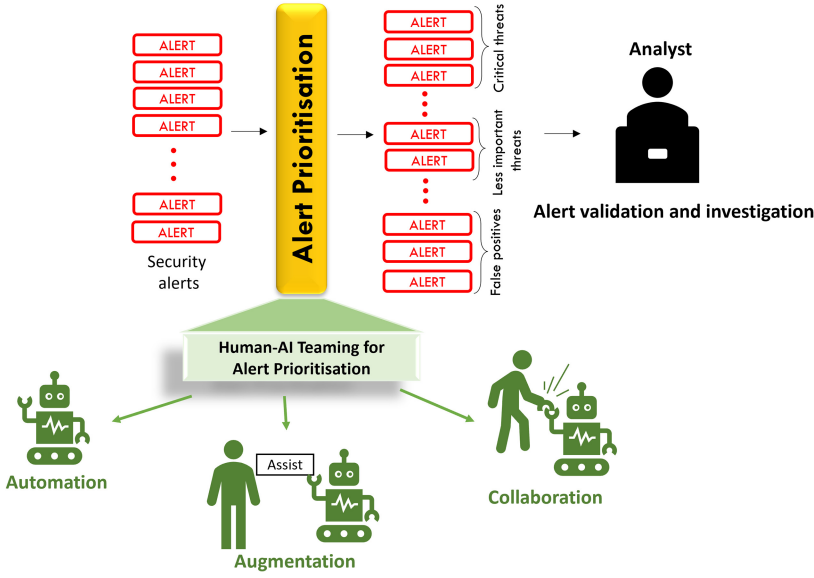


Fig. 1. HAT to support three distinct approaches for AP in SOC: automation, augmentation and collaboration.

- *Automation*: Automation enhances analysts' capabilities by providing insights and explanations for more informed decisions. In automated AP, the AI performs initial AP and provides explainable results for analysts to review and validate. The AI employs specific criteria and methods to conduct the initial AP, while analysts leverage their tacit and contextual knowledge to revise the AP results during the validation phase. The main drawback of this approach is the inefficiency caused by the duplication of effort on AP, which can result in delays, inconsistencies and a higher risk of missed critical alerts.
- *Augmentation*: Augmentation enhances analysts' capabilities by providing insights and explanations for more informed decisions. In augmented AP, the AI performs initial AP and provides explainable results for analysts to review and validate. The AI employs specific criteria and methods to conduct the initial AP, while analysts leverage their tacit and contextual knowledge to revise the AP results during the validation phase. The main drawback of this approach is the inefficiency caused by the duplication of effort on AP, which can result in delays, inconsistencies and a higher risk of missed critical alerts.
- *Collaboration*: In this case, analysts and AI collaborate to perform AP. Collaboration is facilitated through an iterative, sequential and dynamic process. This process involves analysts and AI performing AP sequentially and engaging in iterative exchanges of results. Through this iterative exchange, both analysts and AI learn from their experiences, leading to continual improvement in AP over time. Both AI and analysts have the flexibility to use their own *criteria* and *method* for AP. For instance, in a sequential and iterative process, AI performs AP and sends the prioritised alerts to analysts. Next, analysts validate the priorities, leveraging their expertise and experience not readily available to AI. Analysts then provide feedback to AI on its AP results to improve AP in next iterations, covering their preferences for AP. This iterative process fosters a dynamic collaboration between analysts and AI to leverage their respective strengths and enhance AP accuracy and effectiveness over time.

3 Review Method

We use Wolfswinkel et al.'s [136] method to review the literature on AP. To ensure a thorough review, we repeat the five phases of *define*, *search*, *select*, *analyse* and *present*, in an iterative fashion. This method has been employed by various literature review articles including [58, 93, 104].

- *Define*: This phase clarifies the scope of the article. The objective is to develop a taxonomy for the criteria and methods used for AP, examined through the HAT lens.
 - *AP Criteria Categorisation*. To establish a taxonomy for AP criteria, we first reviewed several recent papers on AP in SOC and identified the main categories of AP criteria in these

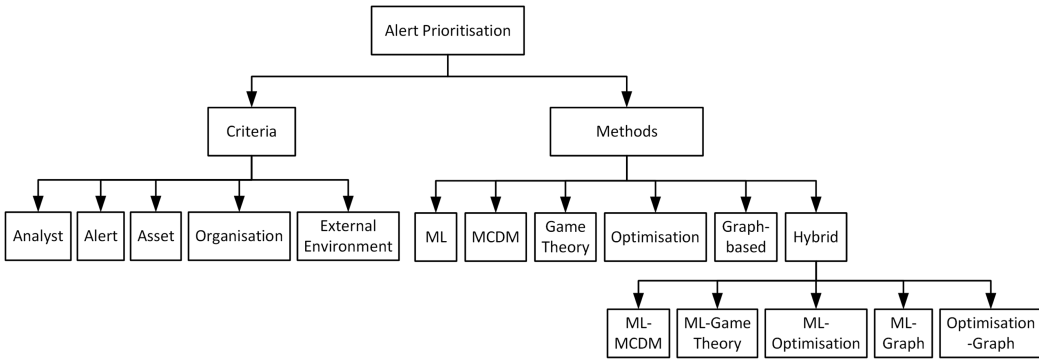


Fig. 2. Alert prioritisation criteria and methods taxonomy.

Table 1. Terms Associated with Search Keywords

Keyword	Subject	Terms
keyword1	AP	alert prioritisation,* alert ranking, threat prioritisation, threat ranking, alarm prioritisation, alarm ranking
keyword2	SOC	security operations centre, cybersecurity operations centre, security intelligence centre, network operations centre, information security operations centre
keyword3	Criteria	criteria, metrics, index, factors
	Method	method, technique, approach, algorithm
keyword4	HAT	human-AI teaming, human machine teaming, human-AI collaboration

*We used both British and American spellings in our search.

papers, serving as our initial categories for AP criteria. Then, we iteratively refined and updated the categorisation based on our findings from the reviewed papers. This resulted in the AP criteria taxonomy presented in Figure 2 and explained in Section 4.

- *AP Method Categorisation.* To develop a taxonomy for AP methods, we first examined several recent papers to find the popular AP methods, providing our initial categories for AP methods. Then, we continuously updated the method groups as we reviewed additional papers. This resulted in the AP method taxonomy shown in Figure 2 and discussed in Section 5.
- *Selected Databases.* We searched well-known scientific databases including IEEE Xplore, ACM Digital Library, Science Direct, Scopus and Google Scholar to find relevant papers.
- *Search Keywords.* We used {keyword1 AND keyword2 AND keyword3 AND keyword4} as the rule to search for the keywords in the documents. Table 1 lists the search terms used.
- *Inclusion/Exclusion Conditions.* We included a paper in our analysis if it discussed (1) AP methods and/or (2) AP criteria, in the context of SOC, while excluding papers that discussed AP in other cybersecurity domains, such as software system security [50] and facility security [134].
- *Search:* We iteratively utilised the keywords and the search rule from the previous phase to search the databases. Initially, all four keywords were used for the database search. We gradually reduced the keywords, first to keyword1, keyword2 and keyword3, and then to just keyword1 and keyword2. The resulting papers were then assessed for relevance to the study, resulting in a total of 647 papers.
- *Select:* We carefully reviewed the 647 papers, sequentially examining their titles, abstracts, introductions and full texts to determine if they met the inclusion/exclusion conditions. As a result, we found 63 relevant papers to our study. To ensure comprehensive coverage, we used the snowballing technique on recent papers published since 2019 to identify any missed

important studies. Applying the same inclusion/exclusion criteria, we found 26 new papers, bringing the total number of papers reviewed in this study to 89. This rigorous review process ensures the comprehensive analysis of relevant literature in our study.

- *Analyse*: We analysed the selected papers to address our research questions. To answer the first research question, we organised and grouped the identified AP criteria in the papers according to their common themes or characteristics, resulting in the development of a taxonomy of AP criteria. Similarly, to answer the second research question, we thoroughly examined the AP methods employed in the papers and established a taxonomy of AP methods in SOC. To answer the third research question, we critically examined the identified AP criteria and methods through the lens of HAT, i.e., automation, augmentation and collaboration, to find the strengths and limitations of existing studies in addressing AP in SOC considering HAT. Based on our findings, we provide recommendations and future research directions to enhance AP in SOC by leveraging HAT.
- *Present*: The key findings from the 89 papers are presented and discussed in the following sections.

4 Overview of Alert Prioritisation Criteria

A taxonomy of AP criteria has been developed based on our review, categorising the AP criteria into five broad categories: *Analyst*, *Alert*, *Asset*, *Organisation* and *External Environment*. We did not encounter any AP criteria that fell outside the scope of these categories. Figure 2 shows the taxonomy of AP criteria, summarised in Table 2. The papers are grouped by the criteria type and HAT type they employ. For instance, the first group of 12 papers utilises analyst-related criteria for automated AP, and the second group of 15 papers employs analyst-related criteria for augmented AP.

To explore the connection between the identified AP criteria and the concepts of automation, augmentation and collaboration, we employed the following conditions:

- If AP is automated, where the machine performs AP automatically using certain criteria without the intervention of analysts, those criteria are categorised under automation.
- If AP is augmented, the machine applies specific criteria to conduct AP and provides explainable results to assist the analysts in validating its prioritisation. Subsequently, during the alert validation phase, analysts may apply additional criteria to refine AP.
- In cases where analysts and the machine collaborate in performing AP, employing their respective criteria, the criteria used by both parties are categorised under collaboration.

It is worth noting that, in augmentation and collaboration scenarios, analysts and the machine may have shared criteria, as well as utilise unique criteria individually. The subsequent sections provide reviews of AP criteria categories. The discussion on AP criteria is given in Section 6.

4.1 Analyst

Analysts play a key role for organisational security, handling tasks such as alert monitoring and triage, incident investigation and threat analysis and response [72, 120]. However, managing a high volume of alerts is challenging, especially with many being false positives. This can cause alert fatigue, where analysts are overwhelmed by the sheer number of alerts and may miss critical threats. To address this issue and improve the alerting system, several criteria can be incorporated to consider the characteristics of analysts, including their time, workload and fatigue levels.

Several studies incorporate criteria addressing analysts' characteristics such as analyst workload and analyst time into the AP model [4, 84, 86, 120]. For example, Shah et al. [120] use two criteria, namely, percentage of effort allocated by analysts and average time spent by analysts for alert analysis, to consider the constraints on workforce resources in SOC. Liu et al. [84] attempt to reduce analyst time spent on alert investigation through eliminating duplicate tasks and consolidating AP

Table 2. Taxonomy of Criteria Used for AP in SOC

Studies	Criteria type	Applied criteria	HAT type
	Analyst Alert Asset Organisation External Environment		Automation Augmentation Collaboration
[4, 41, 62, 84, 86, 110, 120, 135] [45, 59, 101, 119]	•	analyst time, analyst skill, % of effort allocated by analysts, analyst efficiency, stress burnout, fatigue, cognitive workload	✓ × ×
[9, 17, 53, 55, 78, 96, 113, 124] [11, 31, 84, 85, 102, 122, 148]	•	analyst domain knowledge	× ✓ ×
[39, 47, 61, 109, 123, 130, 143] [51, 52, 59, 71, 82, 108, 144]	•	analyst domain knowledge	× × ✓
[3, 8, 14, 18, 89, 99, 105, 119, 140] [20, 25, 45, 57, 59, 100, 116, 137] [26, 97, 110, 120, 131, 139] [6, 7, 67, 91, 112, 114, 117] [46, 60, 73, 74, 79, 83, 127] [10, 23, 70, 94, 132, 133] [12, 30, 38, 56, 141, 142] [5, 44, 62, 110, 135]	•	number of alerts, number of detectors reporting alerts, received at, date, intrusion status CVSS score, hash value, frequency, message ID, email subject, detection time, susceptibility persistence, source, geographical location, host name, device vendor, alert type file size, file type, file name, file category, file detail link, similarity, rareness of events received time, start time, start time integer, incident impact, relationship between alerts time integer, end time, end time integer, destination host, destination IP destination IP integer, source IP, source IP integer, intervals between alerts time since last alert, severity, IP address, exploitability, destination port, event detail link detection ID, signature ID, alert ID, external ID, device external ID, incident ID	✓ × ×
[9, 11, 17, 31, 55, 78, 96, 113, 124, 148] [53, 84, 85, 102, 122]	•	timestamp, source, severity, IP address, detection time	× ✓ ×
[51, 52, 61, 82, 109, 123, 130, 143, 144] [39, 47, 59, 71, 108]	•	impact, destination IP, event URL, severity, geographical location, file size	× × ✓
[23, 30, 59, 60, 70, 73, 83, 120, 139] [3, 6, 7, 10, 44, 86, 110, 137, 142] [12, 45, 74, 99, 109, 135]	•	asset criticality, asset effect, asset dependability, asset maintainability, asset mission confidentiality, integrity, availability, asset control, asset value, asset replaceability configuration management, coverage, asset vulnerability, applicability of attack to asset	✓ × ×
[102]	•	asset criticality	× ✓ ×
[51, 74, 85, 103, 120, 132, 139] [10, 62, 73, 86, 99, 138]	•	cost, number of analysts, analyst cost, business objectives, effectiveness infrastructure compatibility, organisation vulnerability, planning, competitiveness media protection, assurance, reputation, computational resources, timeliness	✓ × ×
[84, 102]	•	computational resources, time efficiency, number of IDS sensors	× ✓ ×
[4, 25, 74, 112, 115–117, 132, 138, 139]	•	context, CTI, attackers' strategies	✓ × ×

outputs into a smaller set for investigation. Huang and Zhu [62] utilise multiple analyst-specific criteria including fatigue, cognitive load, stress and skill to develop a real-time AP. Using these human factors, Huang and Zhu ensure that their AP model is customisable for various analysts.

Table 2 reveals that analyst-specific criteria like analyst time, analyst fatigue and analyst workload, which are mostly used to address analysts' requirements, are only employed by the machine to perform automated AP. A number of reviewed papers address augmentation and collaboration in their AP methods [see, e.g., 51, 124, 144]. An interesting observation is that the sole analyst-related criterion that is employed by the analysts in augmented and collaborative AP is *analyst domain knowledge*. Analyst domain knowledge refers to the specialised expertise and experience that analysts possess, enabling them to understand and interpret complex security incidents, interpret patterns and anomalies and make informed decisions based on contextual information that may not be readily available to the machine. Analysts utilise their domain knowledge to contribute to the AP process through, for example, reviewing and revising the machine-generated AP results and providing feedback to enhance the AP outputs.

4.2 Alert

Alert-related criteria include alert severity, impact, source, and frequency. Our review shows that most studies apply alert-related criteria for AP. Aminanto et al. [8] use 49 alert-related criteria

with categorical (e.g., message ID), numerical (e.g., severity) and timestamp (e.g., receivedTime) values for AP. Message ID is a notification from a security device for alerts; severity is a level of alert risk; and receivedTime is when the alert is received. Alperin et al. [5] apply the geographical location and the severity of alerts to distinguish potential incidents from false positives in AP.

Similar alert-related criteria are employed for automated, augmented and collaborative AP. The main distinction lies in the level of analyst intervention. In automated AP, the machine primarily applies alert-related criteria for AP without analyst involvement. In augmented AP, the machine utilises alert-related criteria for AP and provides explanations and insights about these criteria in determining alert priorities. This helps the analyst make the final AP decision. In collaborative AP, analysts and the machine dynamically interact to do AP using alert-related criteria. In an iterative process, the machine performs AP and analysts review the AP results and provide feedback to the machine by applying their domain knowledge for analysing alert-related criteria [47, 53, 84].

4.3 Asset

In the context of SOC, an asset is any resource, device or system that is of value to the organisation and requires protection from potential security threats.

Various asset-related criteria have been used for AP, as shown in Table 2. For instance, highly critical assets such as data servers require encryption and regular backups. Enterprise software solutions, which need regular updates and support, have significant maintainability costs. Customer databases have high confidentiality and need strict access controls and encryption to prevent unauthorised access. Asset replaceability refers to the capability to substitute an asset considering both cost and time factors. For example, replacing a legacy system involves significant cost and time, making it more difficult to replace. Asset vulnerability determines the asset's exposure to a threat. For example, outdated softwares are highly vulnerable to cyber attacks due to lack of updates.

A comprehensive and consistent AP should consider such asset-specific criteria to assess potential damage to an organisation's important assets [70, 120]. This means that alerts should be prioritised and responded to in a way that minimises security risks to these assets.

As indicated in Table 2, after alert-related criteria, asset-related criteria are the most commonly used criteria for AP in SOC. They are predominantly considered in automated AP. A limited number of studies incorporate asset-related criteria in augmented and collaborative AP [102, 135]. In these cases, the machine employs asset-related criteria for AP, while analysts may utilise asset-related criteria to support their domain knowledge when analysing and adjusting the alert priorities.

4.4 Organisation

Organisations are the users of SOC services. They include businesses, government agencies, financial institutions and other entities that handle sensitive data. The primary function of an SOC is to protect an organisation's information systems and assets against potential security threats. Thus, any reliable AP should cover organisational criteria such as organisational objectives and resources [99, 120]. By considering these criteria, AP can be performed based on the potential impact of an attack on the organisation's objectives, operations, resources and reputation.

Several studies like [74, 85, 120, 138] consider organisational factors like cost, number of analysts and computational resources as AP criteria. For instance, Shah et al. [120] aim to achieve optimal AP while considering constraints on budget and the number of analysts. They also include timeliness, referring to timely detection of suspicious activities, as an organisational objective in their criteria for automated AP. This helps ensure that the AP results stay within the organisation's limited resources while allowing for a prompt response to suspicious activities.

Table 2 shows that few studies utilise organisational criteria in AP. Moreover, only two studies [84, 102] address augmented and collaborative AP using organisational criteria. We observe that, in these cases, the machine uses organisational criteria as input data for AP, whereas analysts do not employ them for AP purposes but apply other AP criteria based on their domain knowledge.

4.5 External Environment

The external environment of an SOC encompasses factors outside of the SOC and its parent organisation that impact cybersecurity practices. This includes regulations, industry standards and compliance requirements. For instance, SOC's must adhere to regulations such as the **General Data Protection Regulation (GDPR)** in Europe [65] and the **Health Insurance Portability and Accountability Act (HIPAA)** in the United States [42]. Industry standards like the **Payment Card Industry Data Security Standard (PCI DSS)** [126] enforce secure handling of cardholder information through rigorous access controls and monitoring. Similarly, compliance requirements such as the **Federal Information Security Management Act (FISMA)** [34] in the United States impose controls to ensure financial data integrity and government information security, respectively. Moreover, industry partners, customers, adversaries and threats linked to any **cyber threat intelligence (CTI)** [28], which can impact AP, are also included in the external environment factors. Thus, a comprehensive and effective AP should address criteria related to the external environment.

Alahmadi et al. [4] introduce context, which refers to the knowledge about customers, third parties and external entities required for alert processing tasks such as AP [4]. However, context has not been practically applied for AP.

Rodriguez and Okamura [112] and Samtani et al. [115] incorporate CTI into AP. CTI involves collecting external data, such as CTI reports, forum and social media posts, articles and dark web content, for relevant information on threats [112, 115]. Coupling CTI as an external threat data source with SOC's internal alert data can result in holistic and advanced AP [112, 115].

Several studies like [25, 74, 116] include attackers' strategies, as an external factor to SOC, in the AP method. They discuss that relying solely on pre-determined internal strategies and criteria of the SOC for AP is inadequate, as attackers can predict and exploit these strategies to evade detection. Thus, attackers' strategies should also be considered for AP [25, 74, 116].

Few studies consider external environment criteria for AP, and all focus on automated AP. None of the studies reviewed use external environment criteria for augmented or collaborative AP.

5 Overview of Alert Prioritisation Methods

We have identified six categories of methods commonly used for AP in SOC: *ML*, *MCDM*, *Game Theory*, *Optimisation*, *Graph-based* and *Hybrid* methods, combining other methods (see Figure 2). Table 3 shows how the papers fit into the taxonomy. The papers are grouped based on the unique combination of method type and HAT type. For example, the first and second groups of papers use ML for automated and augmented AP, respectively. To determine whether the identified methods enable automated, augmented or collaborative AP, the following conditions are examined:

- (i) If AP is solely conducted by the machine using a specific method without any analyst involvement, it is classified as automation.
- (ii) If the machine performs AP with certain methods and provides explanations enabling analysts to understand and revise AP results using their domain knowledge, this implies augmentation.
- (iii) If analysts and the machine collaborate to perform AP by leveraging their respective methods, that represents collaboration. As mentioned, the collaboration typically follows an

Table 3. Taxonomy of Methods Used for AP in SOC

Studies	Method type	Applied method	HAT type
	ML MCDM Game Theory Optimisation Graph based		Automation Augmentation Collaboration
[5, 8, 20, 30, 38, 67, 115]	•	learning to rank, unsupervised/supervised deep learning, logistic regression, RF, Bi-LSTM	✓ × ×
[18, 46, 66, 100, 105, 140]		natural language processing (NLP), Bayesian network, K-means, a priori, doc2vec, isolation forest	
[26, 89, 94, 110, 127, 141]		ensemble unsupervised ML, Gaussian mixture model, frequent pattern mining, DBSCAN clustering	
[23, 57, 62, 110, 112, 133, 135]		correlation techniques, density-based spatial clustering, model tree, RL	
[9, 17, 124]	•	unsupervised/supervised deep learning, decision tree, fuzzy logic framework, attentional model	× ✓ ×
[99, 113]		K-nearest neighbors, naive Bayes, linear discriminant analysis, SVM, AdaBoost, explainable AI	× × ×
[39, 47, 109, 123, 130]	•	fuzzy K-means, deep learning, variational Bayes, temporal pattern mining, model tree, AL, RF	× × ✓
[10, 12, 44, 86, 97]	•	AHP, AHP-group decision-making, ANP, fuzzy AHP	✓ × ×
[3, 137]		ordered weighted averaging (OWA), DEMATEL	
[83, 142]		TOPSIS	
[74, 116, 117, 138, 139]	•	game theory	✓ × ×
[45, 119, 120]	•	mixed integer programming, multi-objective optimisation	✓ × ×
[14, 55, 56, 78, 91, 96, 148]	•	provenance graph	× ✓ ×
[61, 143, 144]	•	provenance graph, characteristic constraint graph	× × ✓
[131]	•	unsupervised clustering, OWA	✓ × ×
[52, 70]	•	SVM, AHP, TOPSIS	× × ✓
[25, 132]	•	DRL, adversarial RL, game theory	✓ × ×
[31]	•	deep learning, DistilBERT, game theory	× ✓ ×
[51, 82]	•	contextual representation learning, logistic regression, RF, XGBoost, SVM, deep learning, game theory	× × ✓
[60, 79, 114]	•	DRL, RF, SVM, deep learning, prioritized K-nearest neighbours classifier, quadratic programming	✓ × ×
		naive base feature selection, integer programming, genetic algorithm	
[59, 108]	•	supervised ML, NLP, mixed integer programming, multi-objective optimisation	× × ✓
[11, 53, 122]	•	provenance graph, Bayesian neural network, GSPAN graph mining	× ✓ ×
[6, 7]		Fuzzy logic inference, DBSCAN clustering, XG Boost, logistic regression, deep learning	
[71]	•	provenance graph, AL	× × ✓
[84, 85, 102]	•	provenance graph, Hill Climber metaheuristic algorithm, greedy heuristic	× ✓ ×

iterative, sequential and dynamic approach, where analysts and the machine sequentially perform AP using their criteria and methods and exchange the results to improve AP outcomes over time.

The following sections review the AP methods. Section 7 discusses the reviewed AP methods.

5.1 Machine Learning

ML algorithms improve computer problem-solving by learning from data [21]. The three types of ML are supervised, unsupervised and **reinforcement learning (RL)**. Supervised learning predicts outputs for new inputs based on known inputs and outputs, while unsupervised learning identifies patterns in input data without knowing output values. RL uses an agent to interact with its environment and select actions that maximise reward or minimise penalty [21].

ML methods excel at analysing large volumes of alert data in real time and leveraging the learned criteria to prioritise alerts effectively [13]. Many ML-based studies focus on automated AP, where the entire AP process is performed by ML and the results are delivered to the analysts [see, e.g., 5, 8, 20, 67]. Supervised, unsupervised and RL are used in different ways to automate AP. Ongun et al. [105] automate AP using ensembles of multiple unsupervised ML methods like Isolation Forest and unsupervised deep learning. They calculate anomaly scores of alerts based on 35 alert features used to rank alerts. The **area under the Precision Recall Curve (PR-AUC)** is used as a measure for the method evaluation. Alperin et al. [5] automate AP using natural language

processing and several supervised ML methods. Natural language processing extracts important features from alert data. These features are then utilised by supervised ML methods such as logistic regression and **random forest (RF)** to classify alerts and calculate vulnerability scores for prioritising alerts. The method is evaluated using the **area under the curve (AUC)** metric.

Limited studies address augmented AP [9, 17, 99, 113, 124]. Sovilj et al. [124] employ an unsupervised deep learning algorithm to prioritise alerts based on their anomaly scores. The authors note that relying only on anomaly scores does not provide enough insight into the causes of critical alerts for analysts. To address this, they use an attentional model as an explainable AI module to provide explanations about the AP outputs for analysts. This assists analysts in understanding the reasons behind the alert priorities, re-prioritising alerts if necessary and effectively investigating and responding to potential threats [124]. Likewise, Andresini et al. [9] adopt the attentional model to provide explanations about AP outputs to analysts, helping them in gaining a deeper understanding of the causes of alert priorities, revising AP results and conducting alert investigations. They use a supervised deep learning algorithm as a multi-class classification method to prioritise alerts. Similarly, Sakazi et al. [113] and Newcomb et al. [99] apply an explainable AI module to enable augmented AP. In another approach, Ban et al. [17] utilise a visualisation tool to present interpretable AP results to analysts, thereby facilitating augmented AP.

Few ML studies focus on collaborative AP [see, e.g., 39, 47, 130]. Teoh et al. [130] combine Artificial Intelligence and Analyst Intuition to do AP. In an iterative process, analysts annotate the alert history to assign weights to the alert clusters/features created by a fuzzy K-Means clustering method. Next, a deep learning algorithm uses these features to classify and prioritise alerts, which are then provided to analysts for further investigation and to give feedback on the weights of classified alerts. **Mean absolute error (MAE)** and **root mean squared error (RMSE)** are used for the method evaluation. Finder et al. [47] apply **active learning (AL)** to iteratively select new and unlabeled alerts and send them for analyst labeling. The labelled alerts by analysts are then used to retrain and update the AL model to prioritise alerts while addressing the continuous creation of unknown alerts in SOC. Thus, analysts dynamically collaborate with the AL method for AP. The percentage of identified malicious instances is used for the evaluation of the AL method.

5.2 Multi-Criteria Decision-Making

MCDM, generally, aims to rank alternatives (solutions) based on multiple criteria to choose the best option. MCDM methods usually aggregate criteria and alternative data in a way to compute the rank or priority of alternatives [15, 88].

MCDM methods provide a proper framework for analysing multi-dimensional problems like AP [97]. They allow analysts to consider and weigh multiple criteria when measuring alert priorities. Existing research focuses on using MCDM for automated AP. Anuar et al. [12] use the **analytic hierarchy process (AHP)** to assess the weights of asset-related criteria such as asset criticality, maintainability and replaceability, as well as alert-related criteria such as alert severity, similarity and frequency used to calculate the rank score of alerts based on their risk index. The AP outputs are dynamically updated by re-ranking alerts whenever a new alert is detected. The method is evaluated against **common vulnerability scoring system (CVSS)** and Snort risk indices, and the results are found to be more distinct and practical [12].

Group multi-criteria decision-making (GMCDM) methods allow incorporating preferences from multiple decision-makers with different criteria to establish rankings for alternatives and achieve consensus among the decision-makers. Eren-Dogu and Celikoglu [44] use group AHP and **Bayesian prioritisation procedure (BPP)** for AP. They include judgments of various influential participants, such as analysts and SOC managers, in AP using group AHP. BPP handles incomplete or inconsistent judgments from decision-makers due to limitations in their time, expertise or

information processing abilities. This ensures that the judgments of different decision-makers are properly addressed in AP.

5.3 Game Theory

Game theory applies mathematical methods to model the interactions among multiple decision-makers (players) that can choose from a set of choices to reach desirable outcomes based on their preferences. It can play an important role in AP by modelling interactions among different entities involved in AP. This includes interactions among different SOC analysts, the interactions between analysts and AI and even the interactions between two opposing parties as attackers and defenders (analysts). Through these models, game theory provides a comprehensive and systematic AP strategy that can improve the overall effectiveness of SOC operations.

Most current studies on applying game theory for AP focus on modelling attacker–defender interactions. To illustrate, Schlenker et al. [116, 117] use a Stackelberg game to find optimal alert priorities by analysing attacker and defender strategies. They suggest modelling this interaction with the Stackelberg assumption, where the defender chooses a strategy first. The attacker then monitors and learns the defender’s strategy before selecting their best response. This means the defender chooses an optimal strategy assuming the adversary will select their best response.

The key motivation for modelling defender–attacker interactions is that relying solely on pre-determined strategies to prioritise alerts is insufficient, as smart attackers can predict and exploit these strategies to evade detection or execute less noticeable attacks. Hence, the prioritisation method should consider the attackers’ strategies. Schlenker et al. [116, 117] use two metrics, expected loss from undetected attacks and runtime, to evaluate their proposed game theory method. Similarly, the remaining reviewed papers in the field of game theory [74, 138, 139] apply a Stackelberg game to model defender–attacker interactions in attack events and prioritise alert categories.

5.4 Optimisation

Optimisation methods aim to optimise one or more objective functions given a set of decision variables, which are typically limited by a set of constraints. Various optimisation models like linear programming, integer programming and quadratic programming address different problems. Moreover, exact or heuristic algorithms are applied to find optimal or near-optimal solutions [92].

Optimisation methods can identify optimal AP while considering SOC resource constraints like budget and analyst numbers. Shah et al. [120] propose a two-step framework for AP. First, alert risk scores are computed using a hierarchical process that includes various alert-related criteria like source, severity and persistence. Second, a mixed integer programming calculates alert priorities to minimise risk while considering resource constraints such as analyst numbers and analyst investigation time. The cumulative risk score of investigated alerts and the alert waiting time are used to evaluate the method. Shah et al. [120] discuss that ranking alerts solely by their risk score and addressing alerts from the highest to the lowest risk may lead to the allocation of all SOC resources for investigating a limited number of alerts. Also, resource constraints, such as analyst availability, might be violated if an alert is selected without considering the availability of necessary resources. An optimisation method can effectively alleviate the above concerns in AP [120].

Farris et al. [45] use a mixed-integer multi-objective optimisation algorithm for AP, constrained by available analyst hours. They apply two evaluation metrics: **time-to-vulnerability remediation (TVR)** and **total vulnerability exposure (TVE)**. TVR is the maximum acceptable time for a vulnerability to remain in the system. TVE is the un-remediated vulnerability density per month. Table 3 indicates that existing studies that solely apply optimisation techniques primarily focus on automated AP.

5.5 Graph Based

Graphs efficiently represent complex data relationships with vertices as data points and edges as relationships [121]. Different graphs like value, proximity and provenance graphs serve varied purposes [1, 76, 121]. For example, provenance graphs can link causal events in a security system, regardless of the time between the two events, and provide explanations for analysis [76]. Graph-based methods are effective tools to model the relationships between alerts [55, 56]. Table 3 indicates that all graph-based studies address either augmented or collaborative AP.

Milajerdi et al. [96] use provenance graphs to model the alerts' relationships and dependencies in **advanced persistent threats (APTs)** and rank APTs by the severity of related graphs. They address augmented AP by showing analysts a provenance graph of APTs, clarifying the attack's scope and impact, and allowing alert re-prioritisation if needed. Provenance graphs summarise attackers' actions and visually represent APTs in real time. Thus, they help analysts understand and investigate the root causes of high-ranking APTs and adjust alert priorities based on their domain knowledge. For example, a 10-million-record database can be represented by a graph with only 16 nodes [96]. The method is evaluated using precision, recall, memory use and runtime metrics.

Zhong et al. [143, 144] use graph representation to model logical relationships between analysts' AP tasks. This differs from other studies [14, 55, 78, 96] that use provenance graphs to capture alerts' dependencies. The proposed method traces analysts' tasks during AP and uses this information to train the machine to perform AP tasks done earlier by analysts. This method has four steps: (1) tracing analysts' AP tasks using computer-aided cognitive task analysis, (2) creating characteristic constraint graphs to identify logical relationships between analysts' AP tasks, (3) mining patterns in the characteristic constraint graphs to find features of analysts' AP tasks, and (4) training machines on analysts' AP tasks using these patterns. The traces can be continuously gathered while analysts are working. Feeding analysts' AP tasks to the machine dynamically can make AP more efficient and reduce analyst workload. Machine can apply learned patterns from analysts' AP tasks to perform subsequent AP, letting analysts focus on new and unknown incidents [143, 144]. This is a representation of collaborative AP, where analysts and machine dynamically work together to do AP. The method is evaluated using the amount of false positives and false negatives as two metrics.

5.6 Hybrid Methods

This section discusses hybrid AP methods that combine the methods from Sections 5.1 to 5.5. Table 3 shows that most hybrid methods combine ML with other methods.

ML-MCDM. Grushka-Cohen et al. [52] combine AHP and **support vector machine (SVM)** to enable collaborative AP. While AHP enables systematic identification of analysts' preferences on AP criteria and weights, SVM analyses alert data and analysts' preferences from AHP to rank alerts. In an iterative collaboration, SVM's AP outputs are sent to analysts for investigation and feedback. The analysts' feedback is then used to update their preferences in AHP and retrain SVM for the next AP. The method is evaluated against different approaches on a dataset using the accuracy metric.

ML-Game Theory. Two recent studies [51, 82] integrate ML and game theory for AP, specifically focussing on collaborative AP. Liu et al. [82] propose a human-in-the-loop approach for AP where a contextual representation learning algorithm extracts contexts from alert streams and historical events, which are then converted by analysts into understandable labels. SHAP [87] as a game theory model is used to increase the explainability of extracted contexts for analysts. Finally, supervised ML methods like SVM and XGBoost are trained on labeled data and classify alerts. The method is evaluated using interpretability and detection accuracy metrics. Similarly, Gelman et al. [51] combine ML and the SHAP model for collaborative AP. In a dynamic collaboration, supervised ML methods classify and prioritise alerts using alert features and analysts provide feedback

Table 4. Distribution of Papers Using AP Criteria across Automation, Augmentation and Collaboration

HAT type	Analyst	Alert	Asset	Organisation	External environment
Automation	11%	60%	25%	15%	11%
Augmentation	19%	19%	3%	2%	0%
Collaboration	16%	16%	0%	0%	0%

on AP outputs based on their domain knowledge. SHAP improves the explainability of alert features for analysts. Analysts' feedback is used to retrain the ML models, reducing repetitive work for analysts and improving the efficiency of less experienced analysts who lack proper domain knowledge. The method is evaluated using threat waiting queue time, false positive reduction and alert quantity reduction [51]. Moreover, Chavali et al. [25] and Tong et al. [132] use **deep reinforcement learning (DRL)** to automate AP and game theory to include attackers' strategies into the AP model.

ML-Optimisation. Hore et al. [59, 60] combine ML and optimisation methods for AP considering resource constraints and organisational context. They address collaborative AP, where ML methods (RF and deep learning) calculate alert priorities based on organisational factors like location and existence of valuable assets. Analysts assign weights to these factors and periodically update them by analysing AP outputs and giving feedback to retrain the ML model. Next, an integer programming algorithm selects high-priority alerts for analyst investigation considering analyst availability and skills constraints. In a completely automated AP [60], first a DRL agent performs resource allocation and then an integer programming model prioritises vulnerabilities, considering the amount of resources allocated by DRL.

ML-Graph. ML algorithms and provenance graphs are integrated for AP [11, 53, 122]. Anjum et al. [11] build APT provenance graphs and feed them into a Bayesian neural network algorithm for APT classification and prioritisation. As an augmentation approach, the provenance graphs associated with highly ranked APTs are given to analysts for explainability, aiding them in understanding why a sequence of events is regarded as an APT attack. This improves analysts' ability to effectively respond to APTs and empowers them to adjust the priority of APTs based on their domain knowledge if necessary. Likewise, Gupta et al. [53] use provenance graphs as input data for multiple ML algorithms (XG Boost, deep learning) to classify and prioritise alerts.

Optimisation-Graph. Provenance graphs are utilised to prioritise APTs based on criteria such as event rareness and time efficiency, using the Hill Climber optimisation algorithm to optimise criterion weights in the priority score [84, 85].

6 Critical Analysis and Future Research on AP Criteria

This section provides the key aspects of existing AP criteria and the potential future research directions. Table 4 presents the distribution of the reviewed papers employing categories of AP criteria across HAT types. Alert-related criteria are most commonly used (60%), followed by asset-related criteria (25%), for automated AP. Analyst-related criteria are adopted in 19% of papers for augmented AP and 16% for collaborative AP. Moreover, there is a noticeable lack of emphasis on asset, organisation and external environment criteria in augmented and collaborative AP.

6.1 Analyst-related Criteria

The SOC is a socio-technical system consisting of humans, technology and processes, with humans being the most crucial for optimal performance [24]. Yet, few studies use analyst-related criteria in AP, often in an ad hoc and implicit manner. For example, Liu et al. [84] claim that their AP method

enhances analyst time efficiency by removing duplicate tasks, but it only uses alert-related criteria like IP address, without directly including analyst time. This approach may overlook analyst availability, leading to alerts being assigned to unavailable analysts for investigation. Limited studies such as [45, 59, 120] integrate analyst criteria into AP. A potential reason for this under-utilisation can be the tendency to view the AP problem purely through a technical lens, emphasising alert-related criteria while overlooking the crucial role of human analysts in the AP task. Further, the literature lacks clear and explicit definitions of analyst-related factors such as fatigue, stress and workload. For instance, Huang and Zhu [62] attempt to address workload and fatigue in the AP problem, but these factors are not explicitly defined and formalised in their approach. Future research can apply analyst-related criteria for AP and examine its effect on AP performance.

Moreover, the analyst-specific criteria typically applied in the current AP methods, e.g., analyst time, appear to be insufficient. While research acknowledges the importance of other analyst criteria, such as workload, emotional state, stress and fatigue [101], they are rarely included in AP methods [62]. These factors significantly affect analysts' performance in their SOC tasks [41, 101] and should be explored in future research.

Table 4 reveals that analyst-related criteria are used in augmented AP (19%) and collaborative AP (16%). However, the focus is primarily on a single factor, namely, *analyst domain knowledge*. Other criteria such as analyst time, fatigue and workload are not typically used by analysts in augmented and collaborative AP. Incorporating these factors by analysts can enhance AP effectiveness by allocating alerts to analysts who are less fatigued and currently available. The limited adoption of these criteria by analysts might be attributed to several reasons, including the lack of real-time tracking systems to monitor and report these criteria, the complexity involved in integrating such dynamic factors into the AP process, the lack of proper guidelines regarding their consideration and possibly the insufficient awareness or training on the potential benefits of including these human-centric factors by analysts in augmented and collaborative AP. Further research should explore how broadening the scope of analyst-related criteria can improve augmented and collaborative AP.

Moreover, analysts with different expertise levels possess varying domain knowledge, which is typically tacit and not explicitly available to others, such as novice analysts and AI. Extracting and explicitly capturing this domain knowledge could greatly benefit AP. For instance, it could be used to train junior analysts to improve their AP performance. Also, we can equip AI with this valuable knowledge, reducing analysts' workload in investigating and fine-tuning AP results. Future studies can use techniques such as knowledge management [49] and knowledge discovery based on ML [27, 37] to elicit analysts' domain knowledge as criteria used by them for AP to enhance augmentation and collaboration in AP and examine its effect on AP performance.

6.2 Alert-related Criteria

A variety of alert-related criteria have been used for AP as evidenced by several reviewed studies. Aminanto et al. employ 49 alert-related criteria [8], while Ongun et al. utilise 35 criteria [105] for AP. However, a notable gap exists. Currently, there are no standardised guidelines or unified collection of alert-related criteria for selecting alert-related criteria across different AP contexts. For example, the CVSS framework offers a structured approach for evaluating alert severity [5] but lacks a standard for alert features used in AP. To address this, we propose developing a holistic standard for alert-related criteria containing essential alert features. Such a standard would benefit SOC by ensuring consistency and reproducibility in AP research, allowing other studies and organisations to replicate and follow the AP procedure of previous studies, yielding similar results.

Furthermore, current studies predominantly focus on the utilisation of alert-related criteria for automated AP. Although alert-related criteria may seem like technical features better handled by

machines, providing analysts with proper explanations and interpretations of these alert features can enable them to contribute more effectively to AP. There is significant potential for future research to explore how alert features facilitate augmentation and collaboration. One example is researching the appropriate level of explanations for alert severity to help analysts understand the underlying causes of severity and resolve security incidents.

6.3 Asset-related Criteria

An effective and feasible AP should include asset-related criteria to provide better coverage and reduce potential harm to assets. While some studies consider certain asset-related criteria in AP, integrating a broader set of criteria remains underexplored. For instance, while some studies use asset criticality alone, including other criteria like asset maintainability, confidentiality and dependency can provide a more comprehensive AP approach, addressing different aspects of assets.

Table 4 shows that asset-related criteria are primarily used for automated AP and rarely for augmented and collaborative AP. This might be because current studies consider asset-related criteria as technical factors more efficiently captured by machines than analysts. However, while more technical asset-related factors like asset value and asset maintainability seem to be more effectively managed by machines in automated AP, less technical factors such as asset mission and asset vulnerability can be considered by analysts in augmented and collaborative AP. Further, similar to alert-related criteria, giving analysts clear explanations and interpretations of the assets' technical factors can help them contribute more effectively to AP. Extending the application of asset-related criteria in augmented and collaborative AP is a valuable direction for future studies, for example, how asset maintainability can be defined using clear statements or easily understandable and applicable features that can aid analysts in conducting AP.

6.4 Organisational Criteria

Few studies include organisational criteria in AP, resulting in incomplete AP approaches that neglect the comprehensive security needs of the organisation [120, 128]. For example, ignoring organisational constraints like cost in AP may lead to investing all resources in a limited number of high-severity alerts, prioritising them over others and missing less severe but still important alerts. Moreover, selecting a resource-intensive alert without considering available organisational resources could breach constraints on resource availability [59, 120]. Thus, future studies should include organisation-specific criteria in AP methods to improve AP outcomes.

Furthermore, the current literature notably lacks the utilisation of organisation-related criteria in augmented and collaborative AP. One potential collaboration involves analysts considering more subjective organisational factors, such as reputation, during AP, with machines handling the objective organisational criteria such as cost and computational resources. Future research can investigate which organisational criteria require/enable augmentation and collaboration in AP and how they can be incorporated into the AP process.

6.5 External Environment Criteria

External environment criteria are the least utilised in AP in the current studies (see Table 4). The literature lacks proper application of external environment criteria for AP. For example, using CTI and attackers' strategies can significantly enhance AP [74, 115], but they are inadequately used for AP. There are other external factors like compliance requirements and industry standards (as reviewed in Section 4.5) that may affect AP results. A potential reason for the limited use of external environment criteria lies in the challenge of capturing and assessing these criteria due to their inherently soft and subjective nature. For example, regulations, CTI and attackers' strategies often lack single, explicit and formalised definitions, leading to varied interpretations across

Table 5. Summary of AP Criteria-related Limitations and Future Research Directions

Criteria group	Limitations	Future research directions
Analyst	i. Lack of clear definition and formulation	i. Include analyst criteria for comprehensive AP
	ii. Lack of real-time tracking systems and guidelines for analysts to consider a wider range of criteria	ii. Develop clear and explicit definitions of analyst-related criteria
		iii. Use variety of analyst-related criteria in AP specially those linked to consider analyst well-being, e.g., fatigue and stress
		iv. Extend the criteria adopted by the analyst from the sole “analyst domain knowledge” to a wider range of criteria like fatigue
		v. Explicitly identify “analyst domain knowledge” for effective utilisation by AI and novice analysts in augmented and collaborative AP
Alert	–	i. Develop a standard for alert-related criteria ii. Explore how alert-related criteria enable augmented and collaborative AP
Asset	–	i. Include asset-related criteria for a comprehensive AP
	–	ii. Examine how asset-related criteria enable augmented and collaborative AP
Organisation	–	i. Include organisation-related criteria for a comprehensive AP
		ii. Investigate which organisational criteria require/enable augmentation and collaboration in AP
External environment	i. Measuring problem due to the soft and subjective nature	i. Include external environment criteria for a comprehensive AP ii. Examine the suitability of external environment criteria for automated, augmented and collaborative AP

different contexts. Integrating these nuanced concepts as AP criteria by machines is challenging as machines typically require well-defined and quantifiable criteria for effective processing. Given that the current literature mostly focuses on automated AP, external environment criteria have not been adequately adopted. A promising solution to this problem is conducting augmented and collaborative AP, where analysts are responsible for evaluating external environment criteria due to their ability to capture and analyse soft and complex concepts. Moreover, there are standards for capturing CTI in machine-readable formats, such as **structured threat information expression (STIX)** and **trusted automated exchange of intelligence information (TAXII)** [64], but their application in AP remains limited in the current literature. Future studies can apply relevant external environment criteria to achieve more comprehensive AP and assess their impact on AP performance. Moreover, they can explore the suitability of utilising external environment criteria by humans, machines or human–machine teams for automated, augmented and collaborative AP. Table 5 summarises the limitations and future research directions related to AP criteria.

7 Critical Analysis and Future Research on AP Methods

This section discusses the advantages and disadvantages associated with each method category used for AP. Next, challenges and future research directions are elaborated.

Table 6 presents the distribution of reviewed articles using AP methods across automation, augmentation and collaboration. As there is no overlap between methods in each category, the total percentage of method and HAT application is obtained by summing the respective columns and rows. ML is the most popular method category used for AP (42%), followed by the hybrid methods (25%). In contrast, game theory and optimisation have the lowest usage percentage in AP, at 7% and 4%, respectively. However, this does not imply that game theory and optimisation cannot make significant contributions to AP, as we will discuss below.

In the current studies, automation is the most focused area, appearing in 58% of papers, followed by augmentation (24%) and collaboration (18%). Notably, the hybrid methods are most used for

Table 6. Distribution of Papers Using Method Categories across Automated, Augmented and Collaborative AP

HAT type	ML	MCDM	Game theory	Optimisation	Graph based	Hybrid	Total HAT	Methods included in hybrid form
Automation	30%	10%	7%	4%	0%	7%	58%	ML-MCDM, ML-Game, ML-Optimisation
Augmentation	6%	0%	0%	0%	8%	10%	24%	ML-Game, ML-Graph, Optimisation-Graph
Collaboration	6%	0%	0%	0%	4%	8%	18%	ML-MCDM, ML-Game, ML-Optimisation, ML-Graph
Total method	42%	10%	7%	4%	12%	25%		

augmentation (10%) and collaboration (8%) in AP. This serves as evidence that hybrid methods, which combine the strengths of individual methods, hold significant potential for augmentation and collaboration. An interesting observation is that graph-based methods are only employed for augmented and collaborative AP. This indicates the promising potential of graph-based methods in facilitating the collaboration between human analysts and machines in AP.

It is worth noting that MCDM, game theory and optimisation are not independently applied in augmented and collaborative AP in the current literature. However, they are used in hybrid methods, primarily in combination with ML, for augmentation and collaboration. We argue that the standalone MCDM, game theory and optimisation methods may not be effective for augmented and collaborative AP due to limitations, such as their inability to handle large alert volumes. However, when integrated with ML, they can serve as supportive components to enhance the implementation of augmented and collaborative AP by ML.

In analysing AP method categories, it is valuable to compare their practical performance and identify their strengths and weaknesses across HAT types. However, the AP method categories are highly diverse in nature, each employing distinct approaches, algorithms and assumptions tailored to specific scenarios. This inherent diversity makes direct comparison challenging, as each method excels in different contexts and faces unique limitations. For example, ML excels in processing large volumes of alert data and identifying complex patterns, while game theory is more suited for strategic decision-making involving multiple players. Further, access to different datasets, use cases and codes of the wide range of reviewed AP methods is challenging. Thus, we have not directly compared the performance of the AP methods. Instead, we provide a detailed analysis of each method category, highlighting their specific strengths and limitations, to give a comprehensive understanding of how each method can be effectively utilised for AP in different HAT types.

7.1 ML Methods

Table 3 shows that numerous studies employ ML methods for AP, encompassing a broad spectrum of techniques ranging from supervised and unsupervised learning to RL.

ML methods have a significant advantage in processing and analysing the large alert data volumes handled by the modern SOC. This ensures reliable AP by accurately examining alert data, identifying key features and uncovering hidden patterns within data. Also, ML enables classifying alerts based on their importance, facilitating their ordering and prioritisation. In more detail, certain methods like K-means clustering, frequent pattern mining and natural language processing uncover hidden features within the alert data [e.g., 5, 8]. This information is then used as input for other techniques, including unsupervised and supervised ML, to conduct AP. Supervised ML techniques such as RF, logistic regression and supervised deep learning primarily aim to classify alert data and determine their priority based on their class [e.g., 18, 82]. In contrast, unsupervised ML techniques like isolation forest mainly focus on anomaly detection to prioritise alerts based on their anomaly scores [e.g., 8, 127]. ML plays a key role in automating AP by efficiently processing large volumes of alert data. The recent progress in automating AP and relieving analysts' data analysis workload heavily relies on the utilisation of ML to capture and analyse the alert data

generated in SOC. ML methods can continuously learn from historical data and adapt to new and unknown threats and attack patterns. This ongoing learning process can improve AP accuracy over time [47, 144].

Moreover, ML methods have shown high potential in addressing augmented AP using techniques like explainable AI [99, 113] and visualisation [17] as well as collaborative AP using active learning [39, 47] and deep learning [109].

However, ML methods mostly depend on training data, and this reliance can be a drawback if the data is incomplete or biased. To effectively detect new and unknown attacks, ML methods need regular re-training, which can create a vulnerability that adversaries may exploit. For example, if an adversary with system knowledge attempts to insert malicious event traces into the benign retraining dataset, ML methods may take malicious traces as benign, ultimately compromising the security of the organisation's network [11]. To mitigate this, ML experts should carefully examine the data used for training ML methods and identify unknown or suspicious patterns in the data [11].

7.1.1 Future Research Directions. Current studies primarily focus on using ML to automate AP. Limited research has been conducted on exploring augmented and collaborative AP using ML. The lack of adoption may stem from analysts' insufficient expertise in working with ML algorithms, which can be mitigated through targeted training initiatives. For example, training programs can be designed to enhance analysts' skills in understanding the strengths and weaknesses of different ML models and interpreting ML-generated results. Our review indicates that only five papers [9, 17, 99, 113, 124] address augmented AP using ML. However, augmented AP is expected to enhance AP by helping analysts understand the underlying causes of alert priorities and respond more effectively to critical attacks. Existing ML research employs explainable AI [99, 113], specifically the attentional model [9, 124], and visualisation [17] as key enablers for augmented AP. However, the application of these enablers in augmented AP is scarce in the current literature. Future research can focus on promoting augmented AP using attentional model and visualisation. Further, researchers can explore a variety of augmentation enablers for ML methods [80], like **local interpretable model-agnostic explanations (LIMEs)** [111] and AIX360 [35], for application in AP.

Collaborative AP in SOC is scarcely addressed using ML methods like AL [39, 47] and deep learning [109]. These studies emphasise iterative collaboration between analysts and ML systems, where analysts provide feedback to improve AP over time. Future research can apply and extend currently adopted ML methods, such as AL, for collaborative AP. Moreover, other ML techniques that can promote HAT, like RL-based learning to rank algorithms [106, 145] and **RL with human feedback (RLHF)** [77, 146], can be explored for AP in SOC. For example, Deep RL-based learning-to-rank algorithms demonstrate strong performance in ranking alternatives while leveraging HAT in other domains, including web search ranking, recommendation systems and document retrieval [106].

A potential avenue for future research is the application of **large language models (LLMs)** like GPT-3.5 and GPT-4 to enable HAT in AP. LLMs excel in language comprehension and can process text data like e-mails and chat logs and produce human-like text. This proficiency makes them valuable for analysing textual features of alerts and generating AP results that analysts can understand. For example, ChatGPT is applied for anomaly detection in log data, reducing the manual verification efforts required by analysts [81]. Further, LLMs can enhance AP by facilitating conversational interactions between analysts and LLM-powered agents and performing activities such as context enrichment and playbook automation [22, 125]. For instance, Nahmias et al. [98] apply an ensemble of LLMs to detect phishing e-mails based on human-crafted questions. The integration of LLMs for augmented and collaborative AP holds significant promise, given their capacity for intelligent conversations, task adaptation and text analysis.

However, LLMs face several challenges, including hallucinations and outdated knowledge. Hallucinations occur when LLMs produce information that appears credible but is actually incorrect or nonsensical. This issue stems from LLMs' lack of domain knowledge, making it hard for them to answer questions beyond their internal scope. Also, keeping LLMs updated with real-time information is difficult, leading to potentially outdated answers, as the external world evolves continuously [68]. Vector databases efficiently manage and store unstructured data. While LLMs often face limitations due to a lack of domain knowledge that cannot be directly uploaded, this domain knowledge, typically unstructured, can be embedded into vector databases for further retrieval [68]. By combining vector databases and LLMs, vector databases can serve as an external knowledge base offering efficient retrieval of domain knowledge for LLMs, a memory system preserving previous related chat contents for each user's dialog or a semantic buffer [54, 68]. This integration can address the previously mentioned issues with LLMs [68]. Further studies can explore the combination of LLMs and vector databases to enhance AP considering various HAT types. For example, vector databases can incorporate analysts' domain knowledge into LLMs, improving LLMs' ability to comprehend analysts' AP preferences and leading to more effective AP.

7.2 MCDM Methods

Different MCDM techniques including AHP, fuzzy AHP, **analytic network process (ANP)**, **technique for order preference by similarity to ideal solution (TOPSIS)** and **decision-making trial and evaluation laboratory (DEMATEL)** are utilised for automated AP. The key advantage of MCDM methods is their ability to include a variety of criteria in AP. The decision-making matrix used in some MCDM methods is an easy and clear way to structure alert data with various features (criteria) and their respective weights (importance). SOC decision-makers can select a proper MCDM method based on their requirements, including the number and type of alert features in AP and the uncertainties associated with the problem.

Existing studies mostly use MCDM methods suitable for addressing the preferences of a single decision-maker. GMCDM methods can facilitate group decision-making in AP through, for example, aggregating the preferences of multiple SOC managers in AP. In this regard, only Eren-Dogu and Celikoglu [44] apply a GMCDM method as group AHP for AP in SOC.

Furthermore, MCDM methods allow analysts to update the weights of alert features during the AP process, providing feedback to enhance AP. This promotes collaborative AP. Existing literature has not fully explored this advantage for collaborative AP. We have identified one study [52] combining AHP with SVM in a hybrid method for collaborative AP. In an iterative process, analysts' preferences are modelled by AHP and then used for retraining the SVM and updating AP.

A drawback of MCDM methods is their limitation in efficiently processing and analysing certain types of alert data. These methods typically require ordinal criteria, where the values of alternatives can be sorted in a descending or ascending order. Considering alert features like source IP address, destination IP address, alert ID and file name, most of these features that should be considered as MCDM criteria are not ordinal. For example, the source IP address provides a series of digits and dots representing the origin of alerts, which is not ordinal. This makes it challenging to consider the source IP address as a criterion for MCDM methods. To address this limitation, few studies [52, 70] integrate MCDM with ML to leverage ML's ability to handle various types of alert data and learn from historical data. For example, in [52], AHP, as an MCDM method, captures analyst preferences, while SVM, as a supervised ML method, captures alert features that AHP cannot handle in AP. However, these studies are limited in number, leaving ample research opportunities in this area.

7.2.1 Future Research Directions. Future research can explore the potential of combining MCDM with ML, which excels in processing large alert data volumes, learning from historical

data and capturing complex alert-related criteria. This combination can address the MCDM limitations in analysing non-ordinal alert-related criteria for AP while also leveraging the MCDM strengths in including criteria with a small amount of data in AP. This integration also enables the application of MCDM in automated, augmented and collaborative AP. By capturing criteria like asset-related, organisational and external environment factors, MCDM can assist ML in handling these criteria that may have limited data and cannot be properly processed by ML. This allows MCDM to support and enhance ML in automated, augmented and collaborative AP.

Moreover, future studies can take advantage of GMCDM to include multiple decision-makers in HAT for AP. For instance, in typical SOC, where several analysts are involved in alert prioritisation and investigation, GMCDM methods like group TOPSIS can help analysts achieve a consensus on the priority of contentious alerts. This approach addresses situations where analysts may hold differing opinions regarding the importance and priority of specific alerts.

7.3 Game Theory Methods

All studies that consider the game theory method for AP in isolation predominantly focus on automated AP [74, 116, 117, 138, 139]. The Stackelberg game models attacker-defender interactions to determine the priority of alerts, which are then sent to analysts for investigation. Also, some hybrid game theory-ML methods automate AP using the Stackelberg game for modelling attacker-defender interactions [25, 132]. This is a promising benefit of game theory for AP. Since attackers may access defence mechanisms of SOC and launch attacks that evade defenders' examination, incorporating attacker strategies into AP can help mitigate these risks [74, 116, 117]. However, the reviewed papers do not address attacker-defender interactions in augmented and collaborative AP.

Our review indicates that the application of game theory for augmented and collaborative AP is primarily limited to the integration of the SHAP model with various ML methods [31, 51, 82]. SHAP as an explainable AI module provides explanations about important alert features to analysts. However, this does not encompass a genuine application of game theory for modelling the interaction between players in HAT for AP. The primary advantage that game theory can offer in the field of HAT for AP is enabling the modelling and analysis of interactions among various stakeholders, including analysts and AI. For instance, it can answer questions like how should multiple SOC analysts interact with a DRL agent as an AI player to achieve optimal AP performance? Also, what are the optimal risk-attitude strategies of analysts and AI in AP to minimise the overall risk in SOC? The existing literature has not fully utilised game theory to tackle these questions, leaving a significant gap in this area of research.

Game theory models are typically not designed to handle the direct processing or analysis of large alert data volumes for AP. For instance, Laszka et al. [74] restrict their analysis to a random sample of 65 alerts from a dataset containing 863,989 alerts to ensure practical implementation of their game theory model. Moreover, game theory models rely on inputs such as players' behaviours, payoffs and strategic choices. In many cases, such detailed and accurate information may not be available, limiting the model's effectiveness. These may explain the limited adoption of game theory methods for AP in all aspects of HAT—automation, augmentation and collaboration—totalling only 7% as indicated in Table 6. Few studies [25, 132] combine game theory with ML to harness ML's capacity to handle large volumes of alert data and address the drawbacks of game theory. However, further research opportunities remain in this area, as these studies are relatively scarce.

7.3.1 Future Research Directions. Game theory has significant potential for enabling augmentation and collaboration in AP through modelling the interactions between analysts and AI for AP. Various game theory models, such as cooperative (players cooperate to achieve proper

strategies), simultaneous (players select strategies simultaneously) and sequential/dynamic (players dynamically select strategies over time), can model different scenarios of collaboration between analysts and AI for AP. For example, in a sequential collaborative AP, where analysts provide iterative feedback to AI to enhance AP over time, a sequential/dynamic game theory method would be beneficial in modelling this collaborative process. Therefore, future studies can leverage game theory to model and analyse collaboration between analysts and AI in AP and find the most effective collaboration scenarios based on their performance outcomes, such as timely response to critical threats, increased analyst productivity and reduced analyst workload and fatigue.

Moreover, a promising future research direction involves integrating game theory with ML to harness their capability in handling large alert data volumes and learning from historical data. Game theory can also enhance ML by modeling the interactions between analysts and AI for AP.

In addition, current studies mainly ignore attackers' strategies in augmented and collaborative AP. But, as discussed, using only defenders' (SOC) strategies for AP is inadequate, since attackers can predict these strategies to evade detection or execute attacks that are less likely to be examined by defenders [74, 138]. Future research in augmented and collaborative AP can enhance the reliability of AP results by incorporating attacker-defender interactions.

7.4 Optimisation Methods

The main advantage of optimisation methods for AP is their capacity to cover resource constraints in AP, resulting in optimal AP outcomes [59, 120]. SOC resources including budget, skillful analysts and computational resources are limited and may be compromised if alerts are chosen without considering resource availability. Exclusively prioritising and addressing alerts based on their highest criticality, without considering resource constraints, can be a narrow-focussed approach that poses a risk to system security. This approach may allocate a majority of SOC resources towards addressing a limited number of highly critical alerts, potentially causing significant damage [59, 120]. For example, the well-known WannaCry cyberattack on computers with an unpatched Microsoft Windows operating system that cost the world economy billions of dollars because of lower criticality alerts continued to remain unaddressed in the network for a long time [59].

Solely relying on optimisation techniques may not effectively facilitate augmented and collaborative AP. Existing studies predominantly employ mixed-integer programming as an optimisation technique for automated AP [45, 119, 120]. However, the practicality of human interaction with these complex and hard mathematical models is limited, restricting their application for augmentation and collaboration in AP. To address this, several studies successfully combine optimisation with ML [59, 108] and graph-based methods [84, 85] in a hybrid form, demonstrating their potential for augmented and collaborative AP. To illustrate, Renners [108] integrates deep learning with a multi-objective optimisation model for collaborative AP. The ML part collaborates with analysts through iteratively sending AP results to them and updating the AP rules based on analyst feedback. The optimisation part acts as a decision support system, finding the optimal AP policy. Therefore, we propose that the combination of optimisation techniques with ML and graph-based methods can effectively contribute to augmented and collaborative AP.

7.4.1 Future Research Directions for Optimisation Methods. The application of optimisation methods for augmented and collaborative AP is under-explored. Future research can investigate how augmented and collaborative AP can be effectively addressed using optimisation methods. This can be achieved by combining optimisation with suitable ML and graph-based methods.

Moreover, current studies in AP predominantly utilise mixed-integer programming optimisation models. Future research can explore the application of alternative optimisation methods such

as dynamic programming and goal programming in AP while addressing HAT. Dynamic programming is used for prioritisation problems in other contexts like generation expansion planning [36].

7.5 Graph-based Methods

Graph-based methods can effectively enable augmented and collaborative AP. By modelling the complex relationships and dependencies between alerts in a simplified and visual graph representation, graph-based methods help analysts understand the root causes of alerts, visualise connections between alerts, respond to them properly and give feedback to the machine to adjust AP outputs [55, 56]. Further, graphs can model logical relationships between AP activities of analysts. Machines can use this knowledge to imitate analysts' AP activities, reduce analyst workload and improve AP outputs [143, 144]. These capabilities properly facilitate augmentation and collaboration in AP as shown in the current literature.

Furthermore, using a graph representation of alerts offers the advantage of detecting highly critical threats that may go unnoticed if investigated individually, without considering the connections to other related alerts [61, 84]. In other words, graphs assist in identifying the “big picture” of alert groups and uncovering sophisticated and interconnected critical alerts that may appear as low-level events when considered separately [61, 84].

However, due to the high volume of daily alerts in SOCs, graphs can become large and complex, which can make their creation computationally expensive. Also, it can be challenging for analysts to understand and interpret these large graphs [14, 55]. To alleviate this problem, Hassan et al. [55] employ the tactical provenance graph, which reduces the size of classical provenance graphs.

In addition, the reviewed studies employing graph-based methods for AP primarily concentrate on alert-related criteria, potentially overlooking crucial factors such as organisational context and assets. This narrow focus on modelling alert relationships may result in a less comprehensive consideration of all relevant criteria in the AP process.

7.5.1 Future Research Directions for Graph-based Methods. Limited studies fully utilise the advantages of graph-based methods to facilitate augmented and collaborative AP. Future research can explore and harness the capabilities of graph models to promote augmentation and collaboration in AP. Another interesting avenue for future research involves integrating graph-based methods with other techniques, such as MCDM and optimisation, to incorporate additional important criteria such as analyst-related and organisational factors into the AP process. This integration would provide more comprehensive AP results and broaden the scope of graph-based methods.

7.6 Hybrid Methods

Hybrid methods have been mostly discussed in previous sections related to their corresponding methods. In this section, we provide an overall discussion about hybrid methods.

Hybrid methods for AP can effectively combine the strengths of different methods while compensating for their limitations. For example, ML compensates for optimisation's limitation in learning from alert data and uncovering patterns in alerts, while optimisation can enhance ML by addressing resource constraints in AP. However, hybrid methods can introduce complexity and pose challenges in terms of development and maintenance, requiring specialised expertise. Furthermore, a number of recent reviewed studies, focusing on augmented and collaborative AP, employ hybrid methods [see, e.g., 31, 51, 59, 82, 108]. This indicates that combining the strengths of various methods through hybrid approaches is a promising way to foster augmented and collaborative AP.

7.6.1 Future Research Directions. Few hybrid methods exist for promoting augmentation and collaboration in AP. However, there is a good potential to leverage the complementary capabilities of combined methods for promoting augmented and collaborative AP. For example, combining

Table 7. Advantages/Disadvantages of Different Method Categories for AP

Method category	Advantages	Disadvantages/limitations
ML	i. Process large volumes of alert data generated in SOC ii. Learn new and unknown attack patterns iii. Good potential for addressing automated, augmented and collaborative AP	i. Vulnerability due to biased training data
MCDM	i. Incorporate a wide range of criteria into AP ii. Include criteria from multiple decision-makers into AP using GMCDM iii. Address collaborative AP via hybrid integration with ML	i. Ineffectiveness to handle non-ordinal alert-related criteria
Game theory	i. Model attacker–defender interactions for AP ii. SHAP as an explainable module integrated with ML for augmented and collaborative AP ii. Model interactions between analysts and AI for augmented and collaborative AP	i. Inefficiency to handle and analyse large volumes of alert data ii. High dependency on hard-to-obtain information about players’ behaviour, strategies and payoffs
Optimisation	i. Capacity to incorporate SOC’s resource constraints into AP ii. Contribute to augmented and collaborative AP via combining with ML and graph-based methods	i. Limitation in augmented and collaborative AP due to complexity of the models
Graph based	i. Enable augmented and collaborative AP via graph representation ii. Detect highly critical threats based on their connections to other alerts	i. Large and complex graphs cause high computational cost and barrier for analyst understanding ii. Narrowly focus on alert-related criteria and ignore other criteria like organisation and asset
Hybrid	i. Leverage the strengths of other methods and compensate for their limitations ii. Enable augmented and collaborative AP	i. Become complex and challenging for execution

game theory and ML can benefit from game theory’s ability to model interactions between AI and analysts, and ML’s ability to learn from historical data to improve interactions. Future research can explore possible combinations of methods in a hybrid format to address augmentation and collaboration in AP to enhance the capabilities of analysts and AI in achieving improved AP performance.

Table 7 summarises the advantages and disadvantages of different method categories. Table 8 presents the future research directions related to the six AP method categories.

7.7 Intersection of Criteria and Methods for AP

This section highlights the key points regarding the intersection of AP criteria and methods. Figure 3 illustrates the intersection of AP criteria and methods across automation, augmentation and collaboration. We provide the important results obtained from our review in Figure 3. It is evident that all types of criteria and methods are utilised for automated AP. However, this is not the case for augmented and collaborative AP. For example, for collaborative AP, only ML, graph-based and hybrid methods are employed in conjunction with analyst and alert-related criteria. While one might assume that automated AP is the most effective HAT type since it can employ all categories of criteria and methods for AP, we argue that this may not be the case. The misconception arises from the limited attention given to augmented and collaborative AP in current studies, as well as the neglect of incorporating different categories of criteria and methods for these HAT types. Although augmented and collaborative AP have the potential to integrate diverse criteria and methods, there are significant gaps in the usage of different criteria and methods within augmented and collaborative AP. Addressing these gaps should be a focus of future research, as doing so would greatly enhance the effectiveness of employing augmentation and collaboration in AP.

Furthermore, a distinction has been identified between analyst-related criteria for automated AP and augmented and collaborative AP. While methods used for automated AP utilise analyst-related criteria corresponding to factors such as fatigue and skill, augmented and collaborative AP methods incorporate analyst domain knowledge as input to their models to address augmentation and collaboration. These methods may also consider other analyst characteristics in their models, but the primary emphasis is on leveraging analyst domain knowledge for augmentation and

Table 8. Summary of AP Method-related Future Research Directions

Method category	Future research directions
ML	i. Promote augmented AP using techniques like explainable AI (e.g., attentional model, AIX360, LIME) and visualisation ii. Foster collaborative AP using methods like RL-based learning-to-rank algorithms, RLHF and AL iii. Conduct training initiatives to enable analysts working with ML to facilitate augmented and collaborative AP iv. Promote automated, augmented and collaborative AP using LLMs v. Combine vector databases with LLMs to address LLMs' limitations and enable more accurate AP results
MCDM	i. Combine MCDM methods with ML to support ML in automated, augmented and collaborative AP and mitigate MCDM's drawback in handling non-ordinal alert-related criteria ii. Take advantage of GMCDM to include multiple decision-makers in HAT for AP
Game theory	i. Use the potential of game theory in modelling the interactions between analyst and AI to enhance augmented and collaborative AP ii. Combine game theory with ML to compensate for game theory's limitation in effectively handling large volumes of alert data and learning from historical data iii. Enhance reliability of augmented and collaborative AP through incorporating attacker-defender interactions
Optimisation	i. Investigate how augmented and collaborative AP are addressed via combining optimisation with ML and graph-based methods ii. Investigate applying various optimisation algorithms like metaheuristics, dynamic programming and goal programming for AP while addressing HAT
Graph based	i. Promote augmentation and collaboration in AP using the capabilities of graph representation ii. Combine graph-based with MCDM and optimisation to include various criteria like analyst-related and organisational factors into AP and provide more comprehensive AP
Hybrid	i. Explore possible combinations of methods in a hybrid format to foster augmented and collaborative AP

collaboration purposes. It is worth noting that capturing analyst domain knowledge is challenging for machines, which is why automated AP methods do not utilise this criterion.

The key enablers of augmented and collaborative AP methods in the current literature are also shown in Figure 3. Existing studies apply explainable AI, visualisation, graph representation and the SHAP model to address augmented AP. Moreover, collaborative AP is addressed through iterative and dynamic collaboration between analysts and AI using analyst feedback to AI. These techniques leverage analyst domain knowledge to achieve augmentation and collaboration purposes.

8 Additional Future Research Directions

This section provides additional future research opportunities related to AP considering HAT.

- *Alternative approaches to HAT.* We have found that alternative approaches to HAT, such as learning-to-defer [75] (where humans and AI learn to defer decision-making to each other when necessary) and human-AI co-learning (where human and AI dynamically learn from each other's task patterns) [118], are rare in the current AP literature. Future studies can explore applying such alternative approaches to the AP context.
- *HAT suitability.* Determining the optimal HAT type (automation, augmentation and collaboration) for specific scenarios is crucial. For instance, Hore et al. use analyst feedback to train ML for collaborative AP in [59], while in [60], they fully automate AP using DRL and integer programming optimisation. Which approach is superior? We should answer these questions by examining the impact of different HAT types (automation, augmentation and collaboration) on the SOC's AP performance using appropriate performance metrics. These metrics can include operational factors such as timely response to threats and threat reduction as well as analyst-specific factors such as fatigue, workload and stress. Future studies should

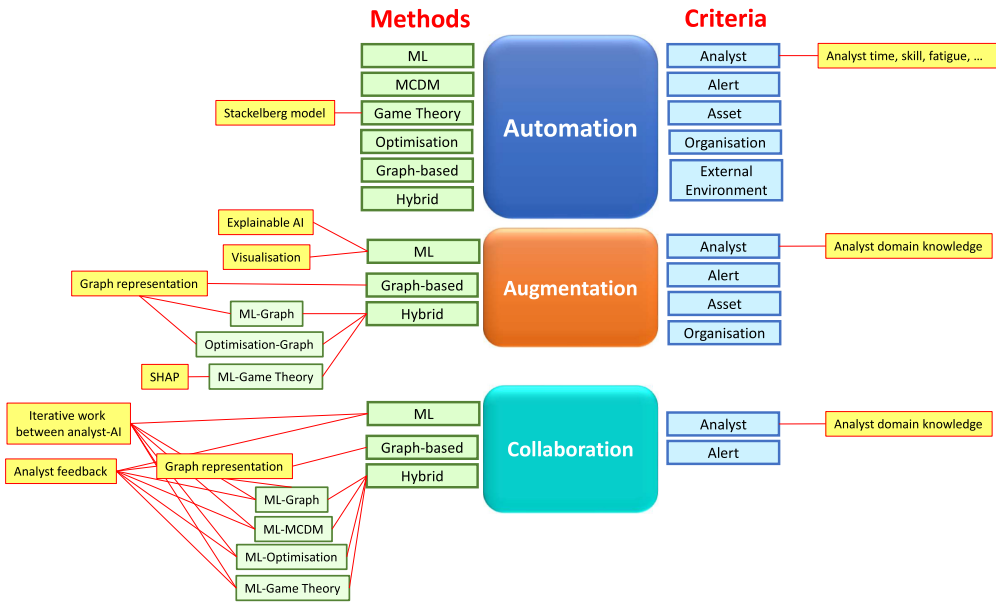


Fig. 3. Intersection of AP criteria and methods across automation, augmentation and collaboration.

Note: In this figure, the green boxes represent the main methods, and the blue boxes denote the main criteria used for AP in relation to automation, augmentation and collaboration. The yellow boxes highlight the key enablers of automated, augmented and collaborative AP methods as discussed in the current literature.

assess different HAT types' suitability for varying SOC situations using appropriate performance metrics. Another potential research direction is finding the optimal balance between automation, augmentation and collaboration for AP in SOC.

- *Standard metrics for evaluating AP performance.* Most studies found in the literature lack standardised performance metrics, hindering comparisons. Establishing standard metrics for evaluating AP results would be beneficial for future research, as it would enable easier comparisons and benchmarking. While method-specific metrics such as precision and recall for ML can still be employed to assess specific aspects of each method, having standardised metrics would enhance the overall evaluation and understanding of AP performance.
- *Practical user studies in AP in SOC.* A promising direction for future research is to conduct user studies and practical experiences for automated, augmented and collaborative AP in SOC based on the identified AP criteria and methods. These studies can involve security analysts and other SOC personnel so that they gain insights into the practical challenges and effectiveness of the proposed AP criteria and methods in real-world HAT scenarios. By gathering empirical data and feedback from users, researchers can refine AP criteria and methods, ensuring they are both theoretically sound and practically applicable in different HAT types.
- *Technical frameworks.* Future research can evaluate the effectiveness of technical tools and frameworks for AP in SOC. For example, the MITRE ATT&CK Framework [16], which provides a comprehensive knowledge base of adversary tactics and techniques, can be integrated to enhance AP systems. Researchers can investigate how aligning AP methods with the MITRE ATT&CK Framework improves the accuracy and relevance of alerts, leading to more efficient incident handling. Additionally, assessing the practical application of this framework through case studies and simulations for automated, augmented and collaborative AP

will help identify best practices and guide future enhancements to AP implementations. The OSC&R [147] Framework is another ATT&CK-like framework targeting supply chains.

9 Challenges for Researchers and Developers

This section outlines some of the main challenges and limitations researchers and developers may encounter in adopting and implementing automated, augmented and collaborative AP.

- **Automated AP**
 - *Adaptability and scalability.* Developing an automated AP system that can adapt to the rapidly changing threat landscape is challenging. Researchers and developers need to continuously update, fine-tune and re-train the system to ensure its relevance and accuracy. Scalability is also a concern, as the system must efficiently process and prioritise a large volume of alerts without compromising performance or accuracy.
 - *Lack of contextual awareness.* Automated systems lack the ability to understand the broader context in which alerts occur. They may not consider factors such as current security posture and recent threat intelligence. This limitation can result in suboptimal prioritisation, ignoring context-sensitive factors. Researchers and developers need to create advanced algorithms that adapt to dynamic contextual information for improved AP.
- **Augmented AP**
 - *Consistency and reliability.* Ensuring consistency in the augmented AP process is challenging. Different analysts may interpret and validate alerts differently, leading to variability in AP outcomes. The reliability of the augmentation approach depends on the analysts' expertise and the quality of the machine-generated results. Poor initial prioritisation by the machine can result in a high cognitive load for analysts, affecting their performance and decision-making.
 - *Explainability and transparency.* For effective augmentation, machine-generated results must be explainable and transparent to analysts. Developing algorithms that can provide clear, interpretable explanations for analyst decisions is a significant challenge.
- **Collaborative AP**
 - *Implementing dynamic interaction.* Implementing an effective interaction between analysts and the machine is complex. It requires seamless integration of human input into the ML process and vice versa. Developing mechanisms for capturing and utilising analysts' feedback to improve the machine's performance over time is challenging and resource intensive.
 - *Availability of analysts.* Conducting research on collaborative AP heavily relies on the availability of analysts. Security analysts are skilled professionals whose time is valuable, making it challenging to run extended experiments involving them over long periods (e.g., multiple days and weeks). However, such long-term experiments are often necessary to accurately capture the real SOC environment.
 - *Trust.* Building and maintaining trust between analysts and AI is fundamental for successful collaboration [69]. Analysts need to trust that AI's recommendations are reliable and that their feedback is valued and incorporated.

10 Conclusion

Effective AP remains a significant challenge for SOC in today's rapidly evolving threat landscape. This article conducted a systematic review, focussing on the criteria and methods employed for AP in SOC through the lens of HAT. Through thorough analysis of existing literature, we identified five distinct categories of criteria and six categories of methods commonly utilised for AP in SOC.

We also examined the advantages and disadvantages associated with different method categories and presented several recommendations for potential research directions.

The key future research directions relating to AP criteria that emerged from this survey are:

- *Analyst*: Develop clear definitions of analyst-related criteria and incorporate them, including those related to analyst well-being such as fatigue and stress. Additionally, explicitly identify and utilise “analyst domain knowledge” for effectively supporting AI systems and novice analysts in augmented and collaborative AP. Also, extend criteria used by analysts from the sole “analyst domain knowledge” to a wider range of criteria.
- *Alert*: Establish a standard for alert-related criteria and explore how these criteria can promote augmentation and collaboration in AP.
- *Asset*: Incorporate asset-related criteria and investigate how these criteria can enhance augmented and collaborative AP.
- *Organisation*: Include organisation-related criteria for a comprehensive analysis and investigate which organisational criteria require or enable augmented and collaborative AP.
- *External environment*: Include external environment criteria and examine their suitability for automated, augmented and collaborative AP.

Similarly, the key future research directions relating to AP method categories are:

- *ML*: Promote augmented AP through techniques like explainable AI and visualisation, and collaborative AP using methods such as RL-based learning to rank algorithms, RLHF and AL. Address automated, augmented and collaborative AP using LLMs and vector databases.
- *MCDM*: Combine MCDM methods with ML to support automated, augmented and collaborative AP to address the drawbacks of MCDM in handling non-ordinal alert-related criteria. Utilise GMCDM for scenarios involving multiple decision-makers.
- *Game theory*: Utilise game theory to model the analyst–AI interactions for augmented and collaborative AP. Combine game theory with ML to handle large volumes of alert data and improve learning from historical data. Enhance reliability of augmented and collaborative AP by incorporating attacker–defender interactions using game theory.
- *Optimisation*: Investigate the combination of optimisation techniques with ML and graph-based methods for augmented and collaborative AP. Explore diverse optimisation algorithms, such as dynamic programming and goal programming, for AP.
- *Graph based*: Leverage graph representation to promote augmented and collaborative AP. Explore graph-based methods with MCDM and optimisation techniques to incorporate diverse criteria for comprehensive AP.
- *Hybrid*: Investigate hybrid methods to enhance augmented and collaborative AP.

We believe that our work serves as a valuable resource for researchers and practitioners working in this field, providing insights and guidance for further advancements. As the threat landscape continues to evolve, we hope that this research will make significant contributions to the ongoing development of more efficient and effective AP strategies.

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